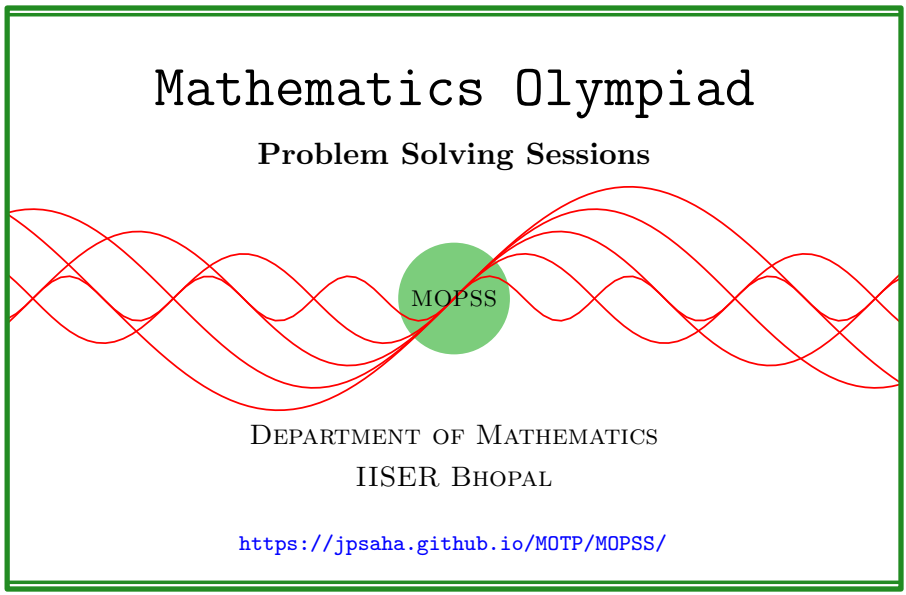
Quadratic polynomials

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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3 Exercise (RMO 1999 P7,) 5
4 Exercise (RMO 2013a P6,) 6

§1 Quadratic polynomials

Exercise 1 (Brazil National Olympiad 2020 Level 2 P2,). The following statement is written on a board:

The equation $x^2 - 824x + \blacksquare 143 = 0$ has two integer solutions.

In the above, $\blacksquare$ represents the first few digits of a number that is blurred. What are the possible equations originally written on the board?

Walkthrough —

(a)

Solution 1. Let α be an integer root of the polynomial $x^2 - 824x + \blacksquare 143$. Note that

$$(\alpha - 412)^2 = 412^2 - \blacksquare 143$$

holds. Note that α is odd. Also note that

$$(\alpha - 412)^2 \equiv 1 \pmod{100}.$$

This implies that

$$|\alpha - 412|^2 \equiv 1 \pmod{25},$$

which yields

$$(|\alpha - 412| - 1)(|\alpha - 412| + 1) \equiv 0 \pmod{25}.$$

Since not both of $|\alpha - 412| - 1$ and $|\alpha - 412| + 1$ are multiples of 5, it follows that $|\alpha - 412|$ is congruent to either 1 or -1 modulo 25. Since $\alpha - 412$ is odd, it follows that $|\alpha - 412|$ is congruent to either 1 or -1 modulo 50. Write

$$|\alpha - 412| = 50k \pm 1$$

for some integer $k \geq 0$. We obtain

$$(50k \pm 1)^2 = 412^2 - \blacksquare 143.$$

Note that

$$412^2 \equiv 800 \times 12 + 144 \equiv 744 \pmod{1000}$$

holds. This gives

$$|\alpha - 412|^2 \equiv 601 \pmod{1000},$$

and hence, we have

$$2500k^2 + 1 \pm 100k = 412^2 - \blacksquare 143.$$

This implies that

$$500k^2 + 1 \pm 100k \equiv 744 - 143 \pmod{1000},$$

which yields

$$500k^2 \pm 100k \equiv 600 \pmod{1000},$$

which is equivalent to

$$5k^2 \pm k \equiv 6 \pmod{10}.$$

This shows that

$$k \equiv \pm 1 \pmod{5}.$$

Since $|\alpha - 412| < 412$, it follows that

$$0 \leq k \leq 8.$$

This implies that $|\alpha - 412|$ is equal to one of

$$50 \pm 1, 50 \times 4 \pm 1, 50 \times 6 \pm 1,$$

that is, $|\alpha - 412|$ is one of

$$49, 51, 199, 201, 299, 301.$$

Note that

$$49^2 \equiv 401 \pmod{1000}, 201^2 \equiv 401 \pmod{1000}, 299^2 \equiv 401 \pmod{1000}.$$

This shows that $|\alpha - 412|$ is one of

$$51, 199, 301.$$

Using

$$\blacksquare 143 = 412^2 - (\alpha - 412)^2,$$

it follows that $\blacksquare 143$ is one of

$$167143, 130143, 79143.$$

This implies that $\blacksquare$ is one of

$$79, 130, 167.$$

Moreover, using

$$79143 = 412^2 - 301^2, 130143 = 412^2 - 199^2, 167143 = 412^2 - 51^2,$$

it follows that the equation $x^2 - 824x + a = 0$ has two integer solutions for any $a \in \{79143, 130143, 167143\}$.

We conclude that the possible values of ■ are $79, 130, 167$. Hence, the possible equations originally written on the board are

$$x^2 - 824x + 79143 = 0, \quad x^2 - 824x + 130143 = 0, \quad x^2 - 824x + 167143 = 0.$$

■

Exercise 2 (RMO 1991 P6,). Find all integer values of a such that the quadratic expression

$$(x + a)(x + 1991) + 1$$

can be factored as a product $(x + b)(x + c)$ where b and c are integers.

Solution 2. Let us establish the following claim.

> **Claim —** A monic quadratic polynomial $f(x)$ with integer coefficients factorizes into the product of two monic polynomials with integer coefficients if and only if the discriminant of $f(x)$ is a perfect square.

Proof of the Claim. Since the discriminant of $f(x)$ is equal to the square of the twice of the difference of its roots, the "only if part" follows.

To prove the "if part", assume that the discriminant of $f(x)$ is a perfect square. Write $f(x) = x^2 + Bx + C$. Note that $B^2 - 4C = n^2$ for some integer n. This shows that the integers B, n are of the same parity. It follows that

$$-B + n, -B - n$$

are even integers. This implies that the polynomial $f(x)$ have integer roots, proving the "if part" of the Claim. □

By the above Claim, the given problem is equivalent to determining all the integers a such that the discriminant of

$$(x + a)(x + 1991) + 1$$

is a perfect square, that is,

$$(a + 1991)^2 - 4(1991a + 1) = n^2$$

holds for some integer n.

Let a, n be integers satisfying the above condition. Note that the above can be rewritten as

$$(a-1991)^2 - n^2 = 4,$$

which is equivalent to

$$(a-1991-n)(a-1991+n) = 4.$$

Noting that the integers $a-1991-n, a-1991+n$ are of the same parity, it follows that the above condition is equivalent to $(a-1991-n, a-1991+n)$ being equal to one of $(2,2), (-2,-2)$, which holds if and only if (a,n) is equal to one of $(1993,0), (1989,0)$.

This proves that the integer values of a satisfying the required condition are precisely $1989, 1993$. ■

Exercise 3 (RMO 1999 P7,). Find the number of quadratic polynomials, ax^2+bx+c, which satisfy the following conditions:

1. a, b, c are distinct,
2. $a, b, c \in \{1, 2, 3, \ldots, 1999\}$ and
3. $x+1$ divides ax^2+bx+c.

Solution 3. Note the third condition is equivalent to $a+c=b$. So the given problem is equivalent to finding the number of triples (a,b,c) where a, b, c are distinct integers lying between 1 and 1999, and satisfy $a+c=b$. Note that it is equal to the number of pairs (a,c) of distinct integers a, c lying between 1 and 1999, such that their sum $a+c$ is at most 1999. Observe that these are precisely the pairs of the form (a,c) where a, c are integers satisfying $1 \leq a \leq 1998$, $1 \leq c \leq 1999-a$ and $a \neq c$. Note that the number of such pairs is equal to

$$\sum_{i=1}^{1998} (1999-i) - 999,$$

where the sum accounts for the pairs (a,c) satisfying $1 \leq a \leq 1998$ and $1 \leq c \leq 1999-a$, and the term 999 accounts for the pairs (a,c) satisfying these two conditions and the condition $a=c$. We conclude that the number of the quadratic polynomials, satisfying the given conditions, is equal to

$$\begin{aligned}\sum_{i=1}^{1998} (1999-i) - 999 &= \sum_{i=1}^{1998} i - 999 \\ &= 999 \times 1999 - 999 \\ &= 1998000 - 1998 \\ &= 1996002.\end{aligned}$$

■

Exercise 4 (RMO 2013a P6,). Suppose that m and n are integers such that both the quadratic equations $x^2 + mx - n = 0$ and $x^2 - mx + n = 0$ have integer roots. Prove that n is divisible by 6.

Solution 4. Since the given quadratic polynomials have integer roots, their discriminants are a perfect squares. It follows that there are nonnegative integers a, b satisfying $m^2 + 4n = a^2, m^2 - 4n = b^2$. This gives

$$2m^2 = a^2 + b^2, 8n = a^2 - b^2.$$

Note that if 3 divides m, then 3 divides a and b, and hence, 3 divides n. Also note that if $m \equiv \pm 1 \pmod 3$ holds, then a^2, b^2 are congruent to 1 modulo 3, and hence, 3 divides n.

If m is an odd integer, then a is odd, and hence, m^2, a^2 are congruent to 1 modulo 8, which gives that 8 divides $4n$, implying n is even. If m is even, then so are the integers a and b, and we have

$$2n = \left(\frac{a}{2}\right)^2 - \left(\frac{b}{2}\right)^2,$$

which implies that the integers $a/2, b/2$ are of the same parity, and hence, 4 divides $(a/2)^2 - (b/2)^2$, implying that n is even.

We conclude that n is divisible by 6. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)